PROJECT SYNOPSIS

Online Voting System

WEB DEVELOPMENT • INFORMATION TECHNOLOGY • INTERNSHIP 2024–27

Project Title	Online Voting System
Domain	Web Development / Information Technology
Technology Stack	HTML, CSS, JavaScript, PHP, MySQL
Academic Year	2024 – 2027
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1. Problem Statement

Traditional paper-based voting systems face critical challenges including voter fraud, ballot tampering, high administrative costs, slow result tabulation, and limited accessibility for differently-abled or geographically remote voters. Manual processes are error-prone and lack transparency, eroding public trust in democratic processes. There is an urgent need for a secure, transparent, and efficient electronic voting mechanism accessible from any internet-enabled device while maintaining the integrity of every cast vote.

2. Objectives and Scope

- Design and develop a secure web-based voting platform accessible from any device.
- Implement OTP-based authentication to ensure one-person-one-vote integrity.
- Provide an admin panel for election setup, candidate management, and monitoring.
- Display live vote counts and graphical result analytics after poll closure.
- Ensure data privacy and prevent duplicate votes through encrypted storage.
- Generate automated audit logs for transparency and post-election verification.
- Support multiple simultaneous elections (national, state, organizational).

Scope: Targets educational institutions, corporate organizations, and government bodies. The initial release focuses on single-round elections; multi-round and preferential voting will be added in future versions.

3. Methodology

The project follows the **Agile SDLC** with iterative sprints:

- **Requirements Analysis:** Functional & non-functional requirements via stakeholder interviews.
- **System Design:** DFDs, Use Case Diagrams, ER Diagrams, and UI/UX wireframes.
- **Database Design:** Normalized MySQL schema for users, candidates, elections, votes.
- **Front-End Development:** Responsive UI with HTML5, CSS3, Bootstrap 5, JavaScript.
- **Back-End Development:** PHP server-side logic; RESTful APIs for data exchange.
- **Security:** Bcrypt hashing, session management, CSRF tokens, SSL/TLS.
- **Testing & QA:** Unit, integration, UAT, and penetration testing.
- **Deployment:** Apache/XAMPP hosting; future migration to AWS/GCP.

4. Hardware & Software Requirements

Component	Specification
Processor	Intel Core i3 or above (1 GHz+)
RAM	Minimum 4 GB (8 GB recommended)
Storage	50 GB HDD / SSD free space
Network	Broadband Internet (10 Mbps+)
Operating System	Windows 10/11, macOS, Ubuntu 20.04+
Web Server	Apache 2.4 / XAMPP / WAMP
Server Language	PHP 8.1+
Database	MySQL 8.0 / MariaDB 10.6
Front-End Stack	HTML5, CSS3, Bootstrap 5, JavaScript ES6
IDE / Editor	VS Code / PhpStorm / Sublime Text
Version Control	Git & GitHub
Browser Support	Chrome 100+, Firefox 98+, Edge 100+

5. Future Work

- **Blockchain integration** for immutable vote recording and transparency.
- **Biometric authentication** (fingerprint / facial recognition) for voter verification.
- **Mobile application** (Android & iOS) to broaden voter reach.

- Support for **preferential and ranked-choice voting** models.
- **AI-based anomaly detection** to identify suspicious voting patterns in real time.
- **Multi-language support** for diverse linguistic communities.
- **Government API integration** (Aadhaar, DigiLocker) for national-scale use.
- **Offline voting mode** with synchronization once connectivity is restored.

6. Project Schedule (Gantt Chart)

Activity	Wk 1-2	Wk 3-4	Wk 5-6	Wk 7-8	Wk 9-10	Wk 11-12
Requirements & Planning	■					
System & DB Design		■				
Front-End Development		■	■			
Back-End Development			■	■		
Security Implementation				■	■	
Testing & Bug Fixing					■	■
Documentation & Deployment						■

7. Conclusion

The **Online Voting System** addresses a real-world need for secure, transparent, and accessible digital democracy. By leveraging modern web technologies and robust security practices, the system significantly reduces cost, time, and human error associated with traditional elections. Its multi-layer authentication, real-time result visualization, and comprehensive audit trail make it stand out. The modular architecture ensures easy scalability, and planned blockchain and biometric enhancements will position it as a production-ready civic-tech solution, demonstrating full-stack development skills with a security-first engineering approach.

8. References / Bibliography

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